

5.9 Appendices

Appendix 5.A: The Lindeberg–Feller Theorem

In Sec.5.4, we state that the multipath components must fulfill the “central limit theorem” in order to give rise to Rayleigh fading. A variety of different forms of the central limit theorem have been suggested. A detailed investigation in [Taricco 2015] derived conditions that is both necessary and sufficient for the CLT to hold for multipath propagation. Specifically, this paper considers the sum of complex random variables, with

$$\tilde{S}_n = \sum_{i=1}^n a_i \exp [j\varphi_i] \quad (5.73)$$

where the φ_i are uniformly distributed between 0 and 2π . The two key theorems state that

1. Iff

$$\begin{aligned} \lim_{n \rightarrow \infty} \sum_{i=1}^n a_i^2 &\triangleq \sigma_{\tilde{X}}^2 < \infty \\ \lim_{n \rightarrow \infty} \sum_{i=1}^n a_i^{2m} &= 0 \quad \text{for } m = 2, 3, \dots \end{aligned} \quad (5.74)$$

then $\tilde{S}_n$ converges in distribution to a circularly symmetric complex Gaussian random variable with zero mean and variance $\sigma_{\tilde{X}}^2$.

2. $\tilde{S}_n$ converges in distribution to a circularly symmetric complex Gaussian random variable with zero mean and variance $\sigma_{\tilde{X}}^2$ if

$$E [a_i^{2m}] = \mathcal{O}(n^{-m}) \quad m = 0, 1, 2, \dots \quad (5.75)$$

which is implied by a stronger assumption

$$\sup_i a_i = \mathcal{O}(n^{-1/2}) \quad (5.76)$$

with probability 1, where $\mathcal{O}$ denotes the standard big-O notation.

Reference

[Taricco 2015] G. Taricco, ”On the Convergence of Multipath Fading Channel Gains to the Rayleigh Distribution,” in IEEE Wireless Communications Letters, vol. 4, no. 5, pp. 549-552, Oct. 2015.

Appendix 5.B: Derivation of the Rayleigh Distribution

Starting with the Gaussian distribution of the real and imaginary part, the pdf of the variables r and ψ can be computed by means of the “Jacobian” [Papoulis 1991]. The joint probability density function $pdf_{I,Q}(I, Q)$ of I and Q is the product of the pdfs of I and Q , since the two variables are statistically independent:

$$pdf_{I,Q}(I, Q) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{I^2}{2\sigma^2}\right) \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{Q^2}{2\sigma^2}\right) \quad (5.77)$$

Next, we express the transformation of variables in a general form:

$$r = f_r(I, Q) = \sqrt{I^2 + Q^2} \quad \psi = f_\psi(I, Q) = \arctan \frac{Q}{I} \quad (5.78)$$

The inverse relationship, i.e., representing I and Q in terms of r and ψ , reads:

$$I = g_I(r, \psi) = r \cos(\psi) \quad Q = g_Q(r, \psi) = r \sin(\psi) \quad (5.79)$$

The pair of values that are solutions of this system of equations are denoted as $(I_1, Q_1), (I_2, Q_2), \dots$ (note that these pairs are of course functions of r and ψ). In our case there is only a single solution, so that only the pair (I_1, Q_1) exists.⁸ Now the Jacobian J is defined as

$$J(r, \psi) = \begin{vmatrix} \frac{\partial g_I(r, \psi)}{\partial r} & \frac{\partial g_Q(r, \psi)}{\partial r} \\ \frac{\partial g_I(r, \psi)}{\partial \psi} & \frac{\partial g_Q(r, \psi)}{\partial \psi} \end{vmatrix} \quad (5.80)$$

The pdf of r and ψ is computed as

$$pdf_{r,\psi}(r, \psi) = pdf_{I,Q}(I_1, Q_1) |J(r, \psi)| \quad (5.81)$$

Inserting into this equation, we immediately get

$$pdf_{r,\psi}(r, \psi) = \begin{cases} \frac{r}{2\pi\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right) & 0 \leq r < \infty, 0 \leq \psi \leq 2\pi \\ 0 & \text{otherwise} \end{cases} \quad (5.82)$$

⁸For the case of a more general transform, where I_2, Q_2 etc. exist as well, we have to compute $pdf(I_1, Q_1)/|J(I_1, Q_1)| + pdf(I_2, Q_2)/|J(I_2, Q_2)| + \dots$

Appendix 5.C: Derivation of the Level Crossing Rate

To compute the joint pdf of r and $\dot{r}$, we start with the joint pdf of the real and imaginary parts x, y , and their temporal derivatives $dx/dt, dy/dt$. Using the same assumptions as in the previous sections (many multipath components, no LOS), this joint pdf is a four-dimensional Gaussian distribution with the correlation matrix.

$$\mathbf{R} = \begin{pmatrix} \Omega_0 & 0 & 0 & \Omega_1 \\ 0 & \Omega_0 & \Omega_1 & 0 \\ 0 & -\Omega_1 & \Omega_2 & 0 \\ \Omega_1 & 0 & 0 & \Omega_2 \end{pmatrix} \quad (5.83)$$

where

$$\Omega_n = (2\pi)^n \int_{f_c - \nu_{\max}}^{f_c + \nu_{\max}} S(f) (f - f_c)^n df \quad (5.84)$$

are the n -th moments of the Doppler spectra. We then get $r, \dot{r}, \psi, \dot{\psi}$ by transforming the x-y pair, as well as the $dx/dt, dy/dt$ pair, to polar coordinates. In the practically important case that $\Omega_1 = 0$, using the appropriate Jacobian gives:

$$pdf_{r, \dot{r}, \psi, \dot{\psi}}(r, \dot{r}, \psi, \dot{\psi}) = \frac{r^2}{4\pi^2 \Omega_0 \Omega_2} \exp \left[-\frac{1}{2} \left(\frac{r^2}{\Omega_0} + \frac{\dot{r}^2}{\Omega_2} + \frac{r^2 \dot{\psi}^2}{\Omega_2} \right) \right] \quad \text{for } r \geq 0. \quad (5.85)$$

If, for example, the Doppler spectrum has a Jakes shape, then

$$\Omega_0 = 3 \frac{E_0^2}{4} \quad \Omega_1 = 0 \quad \Omega_2 = \frac{1}{2} \Omega_0 \left(2\pi f_c \frac{v}{c_0} \right)^2 \quad (5.86)$$

The $pdf_{r, \dot{r}}(r, \dot{r})$ is obtained by integrating over ψ and $\dot{\psi}$

$$pdf_{r, \dot{r}}(r, \dot{r}) = \frac{r}{\Omega_0} \exp \left(-\frac{r^2}{2\Omega_0} \right) \frac{1}{\sqrt{2\pi\Omega_2}} \exp \left(-\frac{\dot{r}^2}{2\Omega_2} \right) \quad \text{for } r \geq 0. \quad (5.87)$$

Inserting this into Eq. (5.49) gives the LCR:

$$N_R(r) = \int_0^\infty \dot{r} \cdot pdf_{r, \dot{r}}(r, \dot{r}) d\dot{r} = \sqrt{\frac{\Omega_2}{\pi\Omega_0}} \frac{r}{\sqrt{2\Omega_0}} \exp \left(-\frac{r^2}{2\Omega_0} \right) \quad \text{for } r \geq 0. \quad (5.88)$$